

AI CAREERS FOR WOMEN LINKEDIN FELLOWSHIP

Sales Performance and Demand Analysis

Project Domain

Excel / Power BI

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:



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We are grateful to our institution, **Smt. Maniben M.P. Shah Women's College of Arts & Commerce**, for providing a supportive environment and necessary resources. We also thank Kaggle for providing the dataset used in this project, which helped us gain practical experience.

Finally, we would like to thank our family, friends, and classmates for their constant encouragement and support. Their motivation helped us stay focused and successfully complete this project.

ABSTRACT OF THE PROJECT

This project is undertaken as part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program by Edunet Foundation in collaboration with Microsoft, LinkedIn, and SAP.

The project focuses on applying data analytics and business intelligence techniques using Microsoft Excel and Power BI to derive meaningful insights from raw data.

The main problem addressed in this project is the presence of unstructured and inconsistent data, including duplicate records, missing values, and formatting errors. Such issues make data analysis difficult and reduce the accuracy of business decisions. The objective of this project is to clean, transform, and analyze the dataset to ensure reliable and meaningful results.

The dataset used in this project was sourced from Kaggle and processed using Microsoft Excel. Data cleaning steps such as removing duplicates, handling missing values, and standardizing formats were performed. After preprocessing, the cleaned data was imported into Power BI for analysis and visualization.

Interactive dashboards were created in Power BI using charts, graphs, and KPI indicators to represent sales trends, product performance, and key business insights. Additionally, AI tools like ChatGPT were used to generate summaries and enhance the overall analysis process.

In conclusion, this project demonstrates how Excel and Power BI can effectively convert raw data into valuable insights. The results show improved data accuracy and better visualization of business performance. This project highlights the importance of data analytics and modern tools in supporting data-driven decision-making.

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CHAPTER 1

Introduction

This project has been undertaken as part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, which aims to provide practical exposure to students in the fields of Artificial Intelligence, Data Analytics, and Business Intelligence. The program, supported by Edunet Foundation in collaboration with Microsoft, LinkedIn, and SAP, focuses on enhancing real-world skills through hands-on projects. This particular project is based on analyzing sales data using Microsoft Excel and Power BI, which are widely used tools in the data analytics industry.

In today's digital era, organizations generate large volumes of data on a daily basis. However, raw data is often unstructured and contains various issues such as missing values, duplicate entries, and inconsistent formatting. These problems make it difficult to extract meaningful insights and can lead to inaccurate decision-making. Therefore, it becomes essential to clean, process, and analyze the data effectively. This project addresses these challenges by focusing on data cleaning and visualization techniques using Excel and Power BI.

The motivation behind this project lies in understanding how real-world business data can be transformed into useful information. Data analytics plays a crucial role in helping organizations make informed decisions, improve efficiency, and identify growth opportunities. By working on this project, practical knowledge of data handling, analysis, and visualization is gained. It also helps in developing problem-solving skills and understanding the importance of Business Intelligence tools in modern industries.

The main objective of this project is to clean and preprocess the sales dataset using Microsoft Excel and then create interactive dashboards using Power BI. The project aims to identify key trends, patterns, and insights from the data, such as sales performance, top-selling products, and category-wise distribution. Additionally, the project also demonstrates the use of AI tools like Copilot or ChatGPT to assist in analysis and report generation.

The scope of this project is limited to data cleaning, analysis, and visualization of the given dataset. It does not include advanced machine learning models or real-time data integration. The focus is on using Excel and Power BI effectively to convert raw data into meaningful insights. This project serves as a foundation for further learning in the field of data analytics and provides a practical understanding of how Business Intelligence tools are applied in real-world scenarios.

1.1 Problem Statement

In the field of data analytics using Microsoft Excel and Power BI, one of the major challenges is dealing with raw and unstructured data. The main problem addressed in this project is the presence of inconsistent, incomplete, and duplicate data in sales datasets, which makes accurate analysis difficult. When datasets contain missing values, incorrect formats, and redundant entries, it becomes challenging to extract meaningful insights. This leads to inefficiencies in data processing and reduces the reliability of analytical results.

This problem is significant because businesses rely heavily on data-driven decision-making. Inaccurate or poorly processed data can result in misleading insights, which may negatively impact business strategies, sales planning, and overall performance. For example, incorrect sales trends or product performance analysis can lead to poor inventory management, reduced customer satisfaction, and financial losses. Therefore, ensuring data accuracy and proper visualization is critical in real-world business environments.

The primary users affected by this problem include business analysts, data analysts, managers, and decision-makers who depend on data for strategic planning. Organizations in sectors such as

e-commerce, retail, and marketing are particularly impacted, as they deal with large volumes of transactional data on a daily basis. Without proper data cleaning and visualization tools, these users may struggle to interpret complex datasets and make informed decisions.

Although tools like Excel and Power BI are powerful, their effectiveness depends on how well the data is prepared and utilized. Existing approaches often involve manual data cleaning, which is time-consuming and prone to human errors. Additionally, without proper knowledge of data transformation techniques and visualization methods, users may not be able to fully leverage these tools. There is also a lack of integration between raw data processing and meaningful visualization in many cases.

This project aims to address these challenges by focusing on systematic data cleaning using Excel and creating interactive dashboards in Power BI. The scope of the problem is limited to preprocessing and visualizing sales data, without including advanced predictive analytics or real-time data systems. By solving these issues, the project helps in improving data accuracy, enhancing visualization, and enabling better decision-making.

1.2 Motivation

The motivation behind selecting this project comes from the increasing importance of data analytics in today's digital and business-driven world. Organizations across all industries rely on data to make informed decisions, improve efficiency, and gain a competitive advantage. This project provided an opportunity to understand how raw data can be transformed into meaningful insights using tools like Microsoft Excel and Power BI. It is personally meaningful as it helps in developing practical skills that are highly relevant in the current job market.

From a professional perspective, learning Excel and Power BI is essential for careers in data analysis, business intelligence, and management roles. This project allowed hands-on experience in handling real-world datasets, performing data cleaning, and creating interactive dashboards. It helped in building problem-solving abilities and understanding how data can influence business decisions. The knowledge gained through this project can be applied in various industries such as e-commerce, finance, marketing, and retail.

The real-world applications of this project are significant. Businesses can use similar data analysis techniques to monitor sales performance, identify customer preferences, and optimize their operations. For example, analyzing sales trends can help companies in demand forecasting, inventory management, and marketing strategies. The dashboards created using Power BI can provide clear and visual insights that make it easier for decision-makers to understand complex data.

The impact of this project on organizations is that it demonstrates how data-driven approaches can improve decision-making and operational efficiency. By cleaning and organizing data effectively, businesses can reduce errors and increase the accuracy of their analysis. The use of visualization tools like Power BI enhances communication of insights, making it easier for stakeholders to take timely actions.

This project also aligns with modern industry trends in Artificial Intelligence and Data Analytics. With the growing use of AI tools such as Microsoft Copilot and ChatGPT, data analysis has become faster and more efficient. The integration of AI with tools like Excel and Power BI is transforming the way data is processed and analyzed. This project reflects these advancements by incorporating AI-assisted analysis and report generation, making it relevant to current industry practices and future technological developments.

1.3 Objectives

The main objectives of this project are as follows:

1. To collect and understand the structure of the Flipkart sales dataset obtained from Kaggle, including its fields, data types, and overall organization.
2. To clean and preprocess the raw data using Microsoft Excel by removing duplicate entries, correcting inconsistent formats, and handling missing values to ensure data accuracy and reliability.
3. To perform domain-specific data analysis using Microsoft Excel and Power BI in order to identify meaningful patterns, trends, and relationships within the dataset.
4. To design and develop interactive dashboards and visual reports in Power BI that effectively communicate insights such as sales performance, product trends, and category-wise distribution.
5. To utilize AI tools such as ChatGPT to assist in data interpretation, report writing, and generation of insights, thereby improving efficiency and productivity.
6. To analyze key business metrics such as total sales, top-performing products, and customer trends to support better decision-making.
7. To enhance practical knowledge and skills in data analytics, business intelligence tools, and visualization techniques for real-world applications.

1.4 Scope of the Project

The scope of this project is focused on analyzing and visualizing sales data using Microsoft Excel and Power BI. The dataset used in this project consists of Flipkart sales records obtained from Kaggle, which includes information such as product details, categories, sales values, and other related attributes. The project involves working with this dataset to extract meaningful insights and improve data quality through systematic processing.

The tools used in this project are limited to Microsoft Excel and Power BI, as assigned under the AICW Fellowship Program. Excel is primarily used for data cleaning and preprocessing tasks, including removing duplicate records, handling missing values, and correcting data formats. Power BI is used for creating interactive dashboards and visual representations of the data to facilitate better understanding and analysis.

The analysis in this project focuses mainly on data cleaning, data transformation, and visualization. It includes identifying sales trends, analyzing category-wise performance, and determining top-performing products. Additionally, AI tools such as ChatGPT are used to assist in generating insights and preparing the project report. The goal is to demonstrate how raw data can be converted into useful information using business intelligence tools.

However, the project has certain limitations. It does not include real-time data integration, advanced predictive analytics, or machine learning models. The dataset used is static and limited to the available records from Kaggle. The project also does not involve the development of automated systems or deployment of dashboards for real-time business use.

This project is intended as a part of the AICW Fellowship Program to showcase practical knowledge and skills in data analytics. It is designed for learning and demonstration purposes only and does not represent a full-scale production system. Despite its limitations, the project provides a strong foundation for understanding data analysis and visualization using Excel and Power BI.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

In the domain of data analytics using Microsoft Excel and Power BI, several tools and techniques have been developed to handle data cleaning, analysis, and visualization. Microsoft Excel has long been used for data preprocessing tasks such as removing duplicates, handling missing values, and applying formulas for data transformation. Features like Pivot Tables, conditional formatting, and data validation are widely used for basic analysis. Similarly, Microsoft Power BI is a powerful business intelligence tool that provides interactive dashboards, data modeling, and advanced visualization capabilities. Pre-built templates and dashboards in Power BI are commonly used by organizations to analyze sales performance and business trends.

Recent advancements have also introduced AI-assisted tools such as Microsoft Copilot and ChatGPT, which help in automating data analysis, generating summaries, and improving report writing through prompt-based interactions. Research studies and official Microsoft documentation highlight the growing importance of integrating AI with traditional tools to enhance productivity and reduce manual effort.

However, existing approaches often have certain limitations. Excel-based methods can become time-consuming and error-prone when dealing with large datasets. Power BI requires proper data preparation and technical understanding to build effective dashboards. Additionally, many users lack knowledge of integrating AI tools effectively for analysis and reporting. This project addresses these gaps by combining Excel for efficient data cleaning, Power BI for clear visualization, and AI tools for enhanced analysis and documentation, thereby providing a simple yet effective solution for handling real-world data.

2.2 Tools and Technologies Review

This project utilizes three main tools: Microsoft Excel, Microsoft Power BI, and AI Chatbots such as ChatGPT. These tools play a crucial role in data cleaning, analysis, visualization, and report generation.

- Microsoft Excel is widely used for data cleaning and preprocessing tasks. In this project, Excel is used to remove duplicate records, handle missing values, and correct data formats. It also provides features such as Pivot Tables, conditional formatting, and data validation, which help in organizing and summarizing the dataset. Excel serves as the foundation for preparing accurate and structured data before analysis.
- Microsoft Power BI is a powerful business intelligence tool used for data visualization and dashboard creation. It allows users to transform cleaned data into interactive reports using charts, graphs, and KPI indicators. Power BI provides an easy-to-use interface that helps in analyzing trends, comparing performance, and presenting insights in a visually appealing manner. It also supports data modeling and DAX functions for advanced analysis.

- AI Chatbots (ChatGPT) are used in this project to assist in report writing, data interpretation, and generating insights. ChatGPT helps in creating structured content such as abstracts, conclusions, and explanations based on the dataset and analysis. It improves efficiency by reducing manual effort and enhancing the quality of documentation through prompt-based responses.

Together, these tools provide an effective solution for handling data analytics tasks. Excel ensures data accuracy, Power BI enables visualization and insight generation, and ChatGPT enhances reporting and analysis, making the overall process efficient and user-friendly.

2.3 Gaps Identified

In the domain of data analytics using Microsoft Excel and Power BI, several gaps are observed in existing approaches. Most existing literature and solutions focus on complex machine learning models, which are not always suitable for beginners or non-technical users. There is limited emphasis on simple, tool-based solutions using Excel and Power BI for effective data cleaning and visualization.

Additionally, while AI tools like ChatGPT are increasingly used, prompt engineering practices for business data analysis are still evolving and lack standardized guidelines. Many users are not fully aware of how to effectively use AI tools for generating insights and reports.

Another gap identified is the lack of integration between data cleaning, visualization, and AI-assisted reporting in a single workflow. Existing methods often treat these processes separately, which reduces efficiency and increases manual effort.

This project addresses these gaps by combining Excel for data cleaning, Power BI for visualization, and ChatGPT for report generation into a simple, integrated approach that is easy to use and suitable for beginners.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The system design of this project follows a structured approach to transform raw sales data into meaningful insights using Microsoft Excel and Power BI. The overall workflow consists of data input, data cleaning, analysis, and visualization, ensuring a smooth flow from raw data to final output.

- **Data Source / Input:** The dataset used in this project is Flipkart sales data downloaded from Kaggle in CSV/Excel format. It contains multiple records related to products, categories, sales values, and other relevant attributes. This raw dataset serves as the input for the entire analysis process.
- **Data Cleaning:** The initial step involves cleaning the raw data using Microsoft Excel. This includes removing duplicate entries to avoid redundancy, handling missing or null values to

ensure completeness, and correcting inconsistent data formats such as text, numbers, and dates. Data is also standardized to maintain uniformity across all records. These steps improve the quality and reliability of the dataset.

- **Analysis / Processing:** After cleaning, the dataset is analyzed using Excel and Power BI. In Excel, basic analysis techniques such as sorting, filtering, and Pivot Tables are used to understand the data structure. The cleaned dataset is then imported into Power BI, where further analysis is performed using visual tools and measures. AI tools like ChatGPT are also used to generate insights, summaries, and explanations based on the data.
- **Visualization/Output:** The final output of the project is an interactive Power BI dashboard that presents the analyzed data in a visual format. Various charts such as bar graphs, pie charts, and line charts are used to display sales trends, category-wise performance, and top-selling products. KPI cards highlight important metrics such as total sales and performance indicators. Additionally, a detailed report is generated using AI tools, making the results easy to understand and useful for decision-making.

3.2 Modules / Components Used

The project is divided into several key modules, each responsible for a specific stage in the data analytics process. These modules work together to convert raw data into meaningful insights.

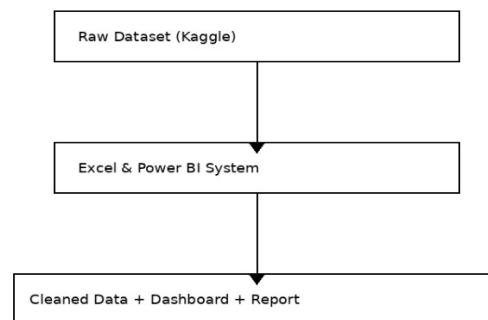
- **Module 1 — Data Ingestion:** This module involves loading the raw Flipkart sales dataset into Microsoft Excel. The dataset, downloaded from Kaggle in CSV/Excel format, is imported and examined to understand its structure, columns, and data types. This step ensures that the data is ready for further processing.
- **Module 2 — Data Cleaning:** In this module, data preprocessing is performed using Microsoft Excel. It includes removing duplicate records, handling missing or null values, correcting inconsistent formats, and standardizing the dataset. This step improves data quality and ensures accuracy in analysis.
- **Module 3 — Data Analysis:** This module focuses on analyzing the cleaned data using Excel and Power BI. Techniques such as sorting, filtering, and Pivot Tables are used in Excel, while Power BI is used for deeper analysis using visual tools and measures. AI tools like ChatGPT are also used to generate insights and explanations based on the data.
- **Module 4 — Visualization / Report Generation:** In this module, interactive dashboards and reports are created using Power BI. Various visualizations such as bar charts, pie charts, line graphs, and KPI cards are used to present key insights. Additionally, structured reports are generated using AI tools to summarize the findings.
- **Module 5 — AI Integration:** This module involves the use of AI tools such as ChatGPT to assist in report writing, data interpretation, and generating meaningful insights. AI helps in improving efficiency, reducing manual effort, and enhancing the quality of analysis.
- **Module 6 — Insight Generation and Decision Support:** This additional module focuses on extracting key findings from the analysis and dashboards. It helps in identifying sales trends, top-performing products, and business patterns, which can support better decision-making.

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram

The DFD Level 0 represents the overall system as a single process. The raw dataset downloaded from Kaggle is taken as input, which is then processed using Microsoft Excel and Power BI. The system performs data cleaning, analysis, and visualization, and produces outputs in the form of cleaned data, interactive dashboards, and analytical reports.



[Insert your DFD Level 0 diagram here.]

DFD Level 1 — Detailed Process Flow

Breaks down the main process into sub-processes. For your project this may include:

- Sub-process 1: Data Loading and Validation

The dataset downloaded from Kaggle is loaded into Microsoft Excel. Basic validation is performed to check column structure, data types, and completeness of the dataset before processing.

- Sub-process 2: Data Cleaning and Transformation

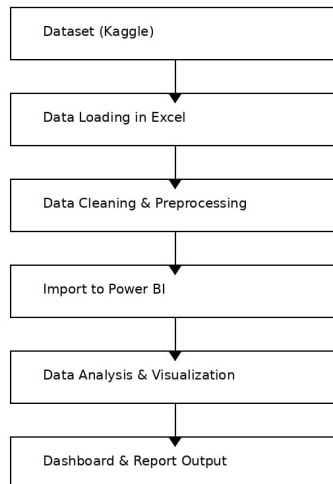
The raw data is cleaned using Excel by removing duplicate entries, handling missing values, correcting inconsistent formats, and standardizing the dataset to ensure accuracy.

- Sub-process 3: Analysis and AI Processing

The cleaned data is analyzed using Excel (Pivot Tables, sorting, filtering) and Power BI. AI tools like ChatGPT are used to generate insights, summaries, and explanations based on the analysis.

- Sub-process 4: Visualization and Report Generation

Interactive dashboards are created in Power BI using charts, graphs, and KPI indicators. Final reports are generated to present the insights in a clear and structured format.



[Insert your DFD Level 1 diagram here showing each sub-process and data stores.]

3.4 Advantages

The proposed approach of using Microsoft Excel, Power BI, and AI tools like ChatGPT offers several advantages in the field of data analytics and business intelligence:

- Automates repetitive data cleaning tasks: The use of Excel functions and structured processes reduces manual effort and minimizes errors during data preprocessing.
- Provides clear and interactive visual insights: Power BI dashboards present data in an easy-to-understand visual format, making it accessible even for non-technical users and stakeholders.
- Enhances efficiency using AI tools: Integration of AI chatbots like ChatGPT helps in faster data interpretation, report generation, and insight extraction, reducing time and human effort.
- Improves decision-making: Accurate data cleaning and effective visualization enable better understanding of trends and patterns, supporting informed business decisions.
- Scalable solution: The same methodology can be applied to larger datasets or different domains such as finance, marketing, and healthcare.
- Cost-effective approach: The project uses widely available tools like Excel and Power BI, which are part of the Microsoft ecosystem and do not require expensive infrastructure.

- User-friendly and easy to implement: The tools used are simple and widely known, making the solution suitable for beginners and non-technical users.
- Flexible and adaptable: The approach can be customized based on different datasets and business requirements, making it versatile for various applications.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

- **Dataset source:** The dataset used in this project was downloaded from Kaggle, a popular online platform that provides real-world datasets for analysis and research. It contains Flipkart sales data in Excel/CSV format, which was used for data cleaning, analysis, and visualization.
- **Issues found:** The dataset contained duplicate records, missing (null) values, and inconsistent data formats in certain columns. Some fields also had incorrect data types and irregular formatting, which affected data accuracy.
- **Cleaning steps applied:** The dataset was cleaned in Microsoft Excel by removing duplicate records, handling missing values, and correcting inconsistent data formats. Data was also standardized and organized to ensure accuracy and consistency for further analysis.
- **Cleaned dataset size:** After data cleaning, the dataset contained approximately 15 rows and 1001 columns, with all duplicates removed and missing values handled. The final dataset was well-structured and ready for analysis in Power BI.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Order ID	Product Name	Category	Price (INR)	Quantity	Total Sales	Order Date	Payment Method	Customer	Month	Year	Profit (INR)	Discount %	Customer Type	Region
2	ORD00001	Education	Books	49069.56	2	98139.12	9/12/2024	Debit Card	3.5	September	2024	19627.82	11	Online	West
3	ORD00002	Jeans	Clothing	12755.85	4	51023.4	12/18/2024	UPI	3.9	December	2024	10204.68	24	Wholesale	North
4	ORD00003	Face Cream	Beauty	11092.19	4	44368.76	5/18/2024	Debit Card	4.4	May	2024	8873.752	19	Online	South
5	ORD00004	Smartwatch	Electronics	32657.81	1	32657.81	11/8/2024	Debit Card	3.2	November	2024	6531.562	15	Online	North
6	ORD00005	Cookware	Home & Kitchen	26590.08	3	79770.24	12/27/2024	UPI	3.1	December	2024	15954.05	12	Retail	North
7	ORD00006	Mixer Grinder	Home & Kitchen	10475.03	2	20950.06	10/22/2024	Wallet	1.7	October	2024	4190.012	25	Wholesale	North
8	ORD00007	Sneakers	Clothing	38647.72	3	115943.2	7/4/2024	Wallet	2	July	2024	23188.63	11	Wholesale	North
9	ORD00008	Table Lamp	Home & Kitchen	24818.59	3	74455.77	12/27/2024	Debit Card	2.4	December	2024	14891.15	30	Wholesale	South
10	ORD00009	Jacket	Clothing	1987.39	2	3974.78	1/18/2025	Debit Card	2.6	January	2025	794.956	23	Online	North
11	ORD00010	Tablet	Electronics	26876.51	2	53753.02	3/14/2024	Debit Card	4	March	2024	10750.6	27	Online	North
12	ORD00011	Perfume	Beauty	46409.57	5	232047.9	2/8/2025	UPI	2.3	February	2025	46409.57	15	Online	West
13	ORD00012	Lipstick	Beauty	48780.78	5	243903.9	4/13/2024	UPI	1.3	April	2024	48780.78	15	Retail	South
14	ORD00013	Education	Books	6670.82	5	33354.1	2/23/2024	COD	1.2	February	2024	6670.82	28	Online	South
15	ORD00014	Smartphone	Electronics	10826.05	1	10826.05	6/2/2024	Wallet	3.7	June	2024	2165.21	25	Retail	East
16	ORD00015	Comic Book	Books	6839.1	1	6839.1	9/12/2024	Credit Card	1.6	September	2024	1367.82	8	Wholesale	South
17	ORD00016	Mixer Grinder	Home & Kitchen	18197.89	5	90989.45	3/30/2024	Debit Card	1.6	March	2024	18197.89	12	Online	South
18	ORD00017	Cookbook	Books	20817.46	2	41634.92	3/30/2024	Debit Card	2.3	March	2024	8326.984	28	Retail	East
19	ORD00018	Perfume	Beauty	44061.25	4	176245	4/22/2024	COD	2.5	April	2024	35249	7	Retail	South
20	ORD00019	Cookbook	Books	4398.59	5	21992.95	3/22/2024	Wallet	4	March	2024	4398.59	26	Online	South

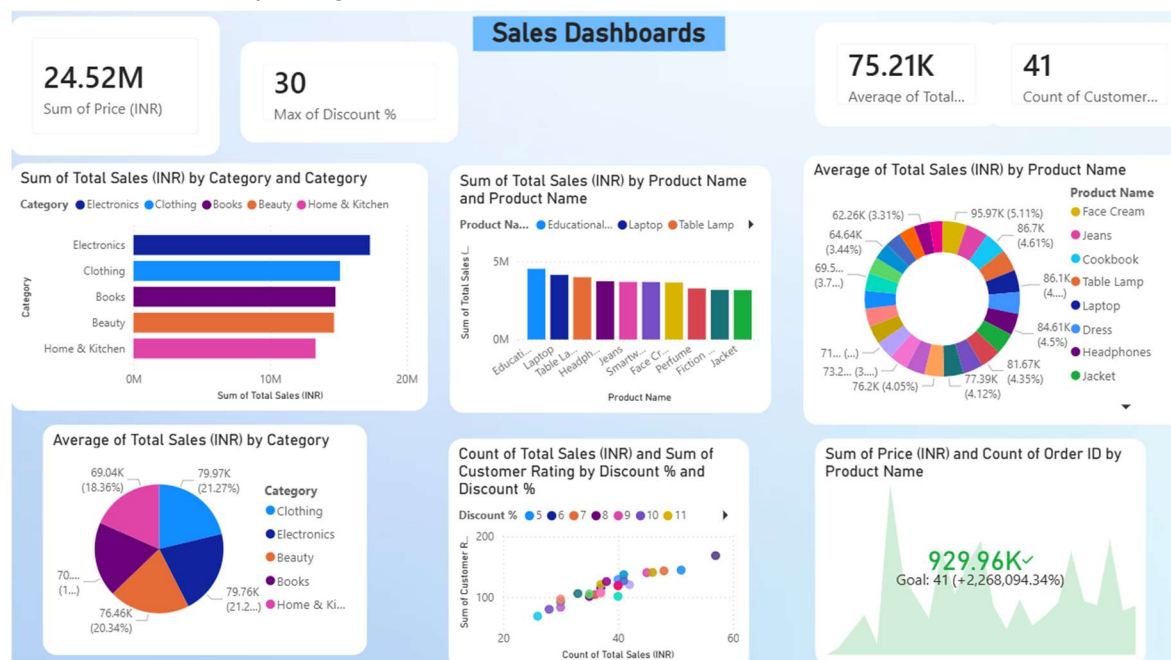
[Insert Figure 1: Screenshot of the raw (uncleaned) dataset here.]

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Order ID	Product N	Category	Price (INR)	Quantity	Total Sale	Order Date	Payment I	Customer	Month	Year	Profit (INR)	Discount %	Customer	Region
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10	ORD00009	Jacket	Clothing	1987.39	2	3974.78	1/18/2025	Debit Card	2.6	January	2025	794.956		23	Online North
11	ORD00010	Tablet	Electronics	26876.51	2	53753.02	3/14/2024	Debit Card	4	March	2024	10750.6		27	Online North
12	ORD00011	Perfume	Beauty	46409.57	5	232047.9	2/8/2025	UPI	2.3	February	2025	46409.57		15	Online West
13	ORD00012	Lipstick	Beauty	48780.78	5	243903.9	4/13/2024	UPI	1.3	April	2024	48780.78		15	Retail South
14	ORD00013	Education Books	Books	6670.82	5	33354.1	2/23/2024	COD	1.2	February	2024	6670.82		28	Online South
15	ORD00014	Smartphone	Electronics	10826.05	1	10826.05	6/2/2024	Wallet	3.7	June	2024	2165.21		25	Retail East
16	ORD00015	Comic Books	Books	6839.1	1	6839.1	9/12/2024	Credit Card	1.6	September	2024	1367.82		8	Wholesale South
17	ORD00016	Mixer Grinder	Home & Kitchen	18197.89	5	90989.45	3/30/2024	Debit Card	1.6	March	2024	18197.89		12	Online South
18	ORD00017	Cookbook	Books	20817.46	2	41634.92	3/30/2024	Debit Card	2.3	March	2024	8326.984		28	Retail East
19	ORD00018	Perfume	Beauty	44061.25	4	176245	4/22/2024	COD	2.5	April	2024	35249		7	Retail South
20	ORD00019	Cookbook	Books	4398.59	5	21992.95	3/22/2024	Wallet	4	March	2024	4398.59		26	Online South
21	ORD00020	T-Shirt	Clothing	32565.77	5	162828.9	1/6/2025	COD	4.4	January	2025	32565.77		25	Wholesale South

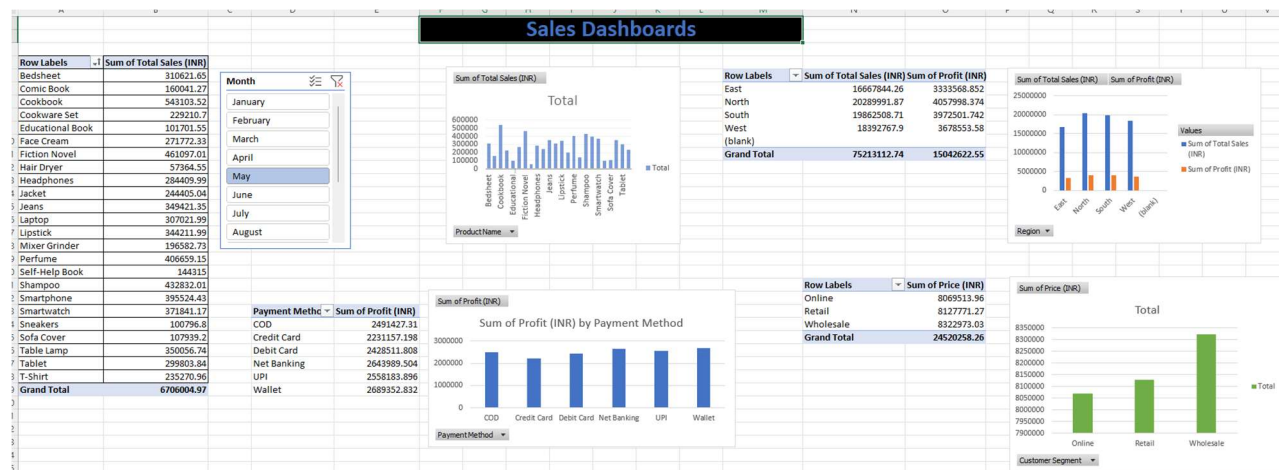
[Insert Figure 2: Screenshot of the cleaned dataset or data validation summary here.]

4.2 Visualization / Output Results

- Dashboard 1: An interactive Excel dashboard using pivot tables and charts to analyze total sales, profit, and performance by product, region, and payment method. It includes slicers for month-wise filtering, making it easy to track sales trends and compare different categories.
- Dashboard 2: A visually appealing Power BI dashboard showing KPIs like total sales, average sales, and order count with advanced charts and insights. It provides dynamic analysis of sales by category, product, discount, and customer ratings with a modern, user-friendly design.



[Insert Figure 3: Screenshot of your main dashboard / report / output here.]

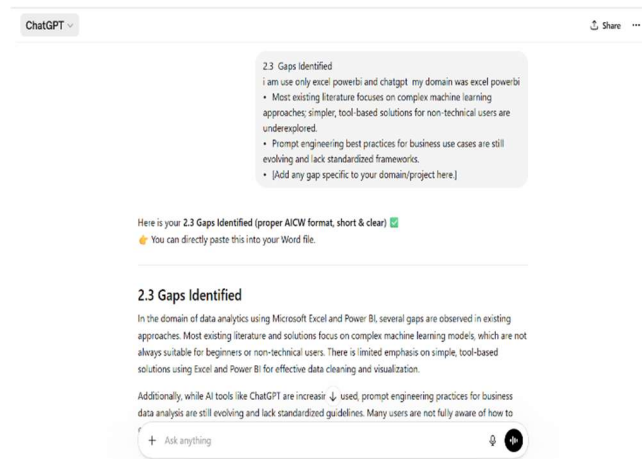


[Insert Figure 4: Screenshot of additional visualization or chart here.]

4.3 AI Tool Integration Results

AI tools played an important role in enhancing the efficiency and quality of this project. ChatGPT was primarily used to assist in report generation, data interpretation, and summarizing insights derived from the dataset.

- **Tool Used:** The main AI tool used in this project is ChatGPT, which helped in generating structured content, explanations, and insights based on the data analysis performed using Excel and Power BI.
- **Prompts Used:** Some of the key prompts used during the project include:
 - "Summarize this sales dataset and identify key trends and insights."
 - "Write a professional report for sales analysis using Excel and Power BI."
 - "Explain the key findings and conclusions from the Power BI dashboard."
- **Output Quality:** The output generated by ChatGPT was clear, well-structured, and relevant to the project requirements. It helped in producing professional-quality content such as abstracts, conclusions, and explanations. The responses were accurate and easy to understand, making them suitable for academic and reporting purposes.
- **How it Enhanced the Project:** The use of ChatGPT significantly improved the efficiency of the project by reducing the time required for report writing and content creation. It helped in organizing ideas, generating insights, and presenting information in a structured format. Additionally, it enhanced the overall quality of the project by providing better explanations and improving the clarity of the analysis. The integration of AI tools made the project more innovative and aligned with modern data analytics practices.



[Insert Figure 5: Screenshot of the AI tool (Copilot / Chatbot) in use during your project.]

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

The project generated several important insights after performing data cleaning, analysis, and visualization using Microsoft Excel and Power BI. The key findings are as follows:

- **Improved data accuracy:** After data cleaning, duplicate records and missing values were removed, which significantly improved the reliability and accuracy of the dataset used for analysis.
- **Clear sales trends identified:** The Power BI dashboard helped in identifying sales trends over time, making it easier to understand patterns in performance and demand.
- **Top-performing products and categories:** A small number of products and categories were found to contribute a major share of total sales, highlighting key revenue drivers for the business.
- **Better visualization of data:** The use of charts such as bar graphs, pie charts, and KPI cards made complex data easier to understand and interpret for decision-making.
- **Efficiency through AI tools:** The use of ChatGPT helped in generating summaries and reports quickly, reducing the time required for documentation and improving overall productivity.
- **Enhanced decision-making support:** The combination of Excel and Power BI enabled the transformation of raw data into actionable insights, which can support better business strategies and planning.

5.2 Project Links

Share the relevant links to your project work for evaluation.

- **GitHub Link of the Project:** [Paste your GitHub repository link here]
- **Video Recording of Project Demonstration:** [Paste your YouTube / Google Drive / OneDrive video link here]
- **Power BI / Dashboard Link (if applicable):** [Paste published dashboard link here]

5.3 Limitations

While the project successfully demonstrates data cleaning, analysis, and visualization using Microsoft Excel and Power BI, there are certain limitations that should be acknowledged.

- **Limited dataset scope:** The project is based on a sample dataset downloaded from Kaggle, which may not fully represent real-world business scenarios. The insights generated are limited to the available data.

- No real-time data integration: The dataset used is static, and the dashboard does not support real-time data updates or live data connections, which limits its applicability in dynamic business environments.
- Lack of advanced analytics: The project focuses only on descriptive analysis and does not include predictive modeling, machine learning, or advanced statistical techniques.
- Dependence on manual data cleaning: Data preprocessing in Excel involves manual steps, which can be time-consuming and may lead to human errors if not handled carefully.
- Limited AI integration: Although ChatGPT was used for report generation and insights, deeper integration with AI tools such as automated data analysis or Copilot features was limited.
- Tool limitations: Power BI dashboards were not deployed online, which restricts accessibility and sharing capabilities for broader users.

5.4 Future Work

The current project provides a strong foundation in data cleaning, analysis, and visualization using Microsoft Excel and Power BI. However, there are several opportunities to enhance and extend this project in the future to make it more advanced and impactful.

- Integration of real-time data: The project can be improved by connecting real-time data sources using APIs or tools like Power Automate. This will allow dashboards to update automatically and provide live insights.
- Application of machine learning models: Advanced techniques such as sales forecasting, trend prediction, and anomaly detection can be applied to the cleaned dataset to generate predictive insights.
- Development of user-friendly applications: A web-based interface or Power Apps solution can be developed to allow non-technical users to interact with the dashboard easily and generate reports without technical knowledge.
- Expansion of dataset: The analysis can be extended by including more diverse datasets such as customer behaviour, marketing data, or inventory data for a more comprehensive business analysis.
- AI-driven automation: A standardized prompt library for tools like ChatGPT can be created to automate report generation, data summarization, and insight extraction more efficiently.
- Cloud deployment and sharing: The dashboards can be published using Power BI Service for online access, collaboration, and sharing with stakeholders across organizations.
- Enhanced visualization features: Additional advanced visuals and custom dashboards can be implemented to improve user experience and provide deeper insights.

5.5 Conclusion

This project successfully demonstrated how raw and unstructured sales data can be transformed into meaningful insights using Microsoft Excel and Power BI. The main problem of inconsistent, incomplete, and duplicate data was addressed through effective data cleaning and preprocessing techniques in Excel. After preparing the dataset, Power BI was used to create interactive dashboards that clearly presented key insights such as sales trends, category performance, and top-performing products.

The project achieved all its objectives, including data collection, cleaning, analysis, and visualization. The integration of AI tools like ChatGPT further enhanced the process by assisting in generating reports and interpreting data insights efficiently. The final output provided a clear and structured representation of the data, making it easier for users to understand and analyze.

The overall impact of this project lies in its ability to improve decision-making through accurate data analysis and visualization. It highlights the importance of Business Intelligence tools in converting raw data into actionable insights. This project can be applied in real-world business scenarios to enhance performance and strategy planning.

From this fellowship experience, valuable practical knowledge was gained in data analytics, visualization techniques, and AI-assisted reporting. It also improved problem-solving skills and understanding of real-world data challenges. Overall, this project adds significant value by demonstrating the effective use of Excel and Power BI in the field of data analytics.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: <https://support.microsoft.com/en-us/excel>

[2] Microsoft Corporation. Power BI Documentation. Available at: <https://learn.microsoft.com/en-us/power-bi/>

[3] Kaggle. *Flipkart Sales Dataset*. Available at: <https://www.kaggle.com>

[5] OpenAI. *ChatGPT Documentation and Usage Guide*. Available at: <https://openai.com/chatgpt>

[6] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.

APPENDICES

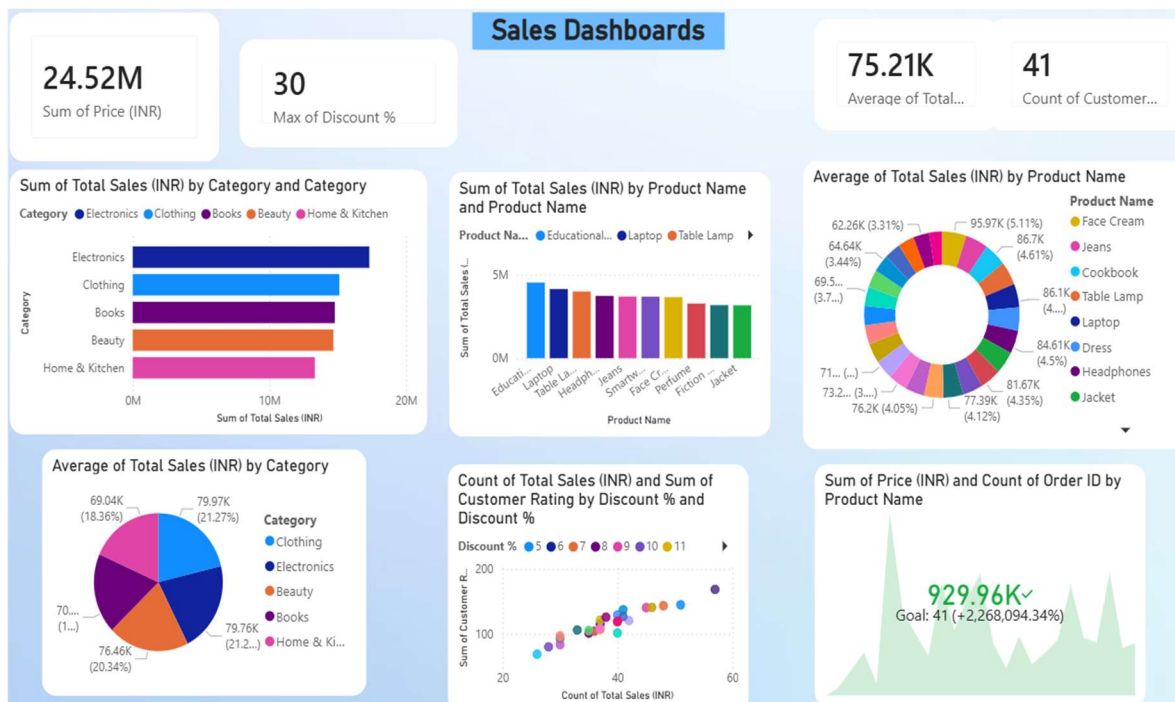
Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

[Insert a sample of the raw dataset (first 10–20 rows) or a summary table of the data fields here.]

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Order ID	Product Name	Category	Price (INR)	Quantity	Total Sales	Order Date	Payment Method	Customer	Month	Year	Profit (INR)	Discount %	Customer	Region
2	ORD00001	Education Books	Books	49069.56	2	98139.12	9/12/2024	Debit Card	3.5	September	2024	19627.82		11 Online	West
3	ORD00002	Jeans	Clothing	12755.85	4	51023.4	12/18/2024	UPI	3.9	December	2024	10204.68		24 Wholesale	North
4	ORD00003	Face Cream	Beauty	11092.19	4	44368.76	5/18/2024	Debit Card	4.4	May	2024	8873.752		19 Online	South
5	ORD00004	Smartwatch	Electronics	32657.81	1	32657.81	11/8/2024	Debit Card	3.2	November	2024	6531.562		15 Online	North
6	ORD00005	Cookware	Home & Kitchen	26590.08	3	79770.24	12/27/2024	UPI	3.1	December	2024	15954.05		12 Retail	North
7	ORD00006	Mixer Grinder	Home & Kitchen	10475.03	2	20950.06	10/22/2024	Wallet	1.7	October	2024	4190.012		25 Wholesale	North
8	ORD00007	Sneakers	Clothing	38647.72	3	115943.2	7/4/2024	Wallet		2 July	2024	23188.63		11 Wholesale	North
9	ORD00008	Table Lamp	Home & Kitchen	24818.59	3	74455.77	12/27/2024	Debit Card	2.4	December	2024	14891.15		30 Wholesale	South
10	ORD00009	Jacket	Clothing	1987.39	2	3974.78	1/18/2025	Debit Card	2.6	January	2025	794.956		23 Online	North
11	ORD00010	Tablet	Electronics	26876.51	2	53753.02	3/14/2024	Debit Card	4	March	2024	10750.6		27 Online	North
12	ORD00011	Perfume	Beauty	46409.57	5	232047.9	2/8/2025	UPI	2.3	February	2025	46409.57		15 Online	West
13	ORD00012	Lipstick	Beauty	48780.78	5	243903.9	4/13/2024	UPI	1.3	April	2024	48780.78		15 Retail	South
14	ORD00013	Education Books	Books	6670.82	5	33354.1	2/23/2024	COD	1.2	February	2024	6670.82		28 Online	South
15	ORD00014	Smartphone	Electronics	10826.05	1	10826.05	6/2/2024	Wallet	3.7	June	2024	2165.21		25 Retail	East
16	ORD00015	Comic Books	Books	6839.1	1	6839.1	9/12/2024	Credit Card	1.6	September	2024	1367.82		8 Wholesale	South
17	ORD00016	Mixer Grinder	Home & Kitchen	18197.89	5	90989.45	3/30/2024	Debit Card	1.6	March	2024	18197.89		12 Online	South
18	ORD00017	Cookbook	Books	20817.46	2	41634.92	3/30/2024	Debit Card	2.3	March	2024	8326.984		28 Retail	East
19	ORD00018	Perfume	Beauty	44061.25	4	176245	4/22/2024	COD	2.5	April	2024	35249		7 Retail	South
20	ORD00019	Cookbook	Books	4398.59	5	21992.95	3/22/2024	Wallet		4 March	2024	4398.59		26 Online	South

Appendix B — Key Excel Formulas / DAX Measures Used



- Power BI DAX: dashboard

Appendix C — AI Prompts Used

New test session
Evaluate
Variables

Which Payment method is most used?

Most Used Payment Method

Based on the available sales data, the most frequently used payment methods are:

- UPI
- Wallet
- Debit Card

Among these, **UPI** appears most often in the recent sales records, making it the most used payment method in your dataset ¹.

If you need a detailed breakdown or a specific report, please let me know!

reference
1
_flipkart_sales.xlsx

New test session
Evaluate
Variables

What is the average of customer rating?

Average Customer Rating

Based on the sales data, the customer ratings for recent orders are as follows:

- 3.5
- 3.9
- 4.4
- 3.2
- 3.1
- 1.7
- 2.0

Calculating the average: $(3.5 + 3.9 + 4.4 + 3.2 + 3.1 + 1.7 + 2.0) / 7 = 3.11$

The average customer rating is approximately 3.1 out of 5 ¹

Ask a question or describe what you need
0/2000

Appendix D — Screenshot Gallery

[Insert any additional screenshots, code snippets, or supplementary visuals here that were not included in the main chapters.]

